Time Series Analysis

Purpose

Examine how COVID-19 VAERS reporting volume, severity rates, and narrative cluster membership shift over the study period (2020–2025). Temporal trends are modelled using ordinary least squares (OLS) linear regression with HC3 heteroskedasticity-robust standard errors.

Analysis pipeline

1. **Setup & Imports**
2. **Load Data**
3. **Monthly Aggregation** — compute monthly report counts and severe-event proportions
4. **Reporting Volume Trend** — Linear regression: monthly report count ~ time
5. **Severity Proportion Trend** — Linear regression: monthly severe proportion ~ time
6. **Cluster Prevalence Over Time** — compute monthly proportions for each narrative cluster
7. **Plot Cluster Prevalence** — visualise how cluster shares change month-by-month
8. **Cluster Trend Regressions** — Linear regression for each cluster's monthly proportion ~ time
9. **Annotated Timeline Plots** — overlay key vaccine rollout phases on reporting-volume trend

1. Setup & Imports

In [ ]:
```
#Setup
import sys
import warnings

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
from statsmodels.tsa.seasonal import seasonal_decompose
from scipy import stats

warnings.filterwarnings("ignore")

print("Python executable:", sys.executable)
print("Python version:", sys.version.split()[0])
print("Pandas version:", pd.__version__)
```

```
Python executable: /Users/ariellerothman/Desktop/Sperm Cell Analysis/.venv/b
in/python
Python version: 3.12.10
Pandas version: 2.2.3
```

2. Load Data

Load the clustered VAERS dataset produced by `clustering_code.ipynb`. This file contains all cleaned, engineered, and cluster-labelled records and is the sole input for this notebook. `OUTDIR` is set here so all output files and saved figures are written to the shared `Outputs/` folder.

In [ ]:
```
# Load clustered df_clean directly
from pathlib import Path

HERE   = Path().resolve()  # project root (the folder containing this notebo
OUTDIR = str(HERE / "Outputs")
os.makedirs(OUTDIR, exist_ok=True)

df_clean = pd.read_csv(HERE / "Outputs" / "df_clean_with_clusters.csv")

print(f"✓ Loaded df_clean with clusters: {df_clean.shape}")
print(f"  Clustered rows: {df_clean['CLUSTER'].notna().sum():,}")
```

```
✓ Loaded df_clean with clusters: (986096, 106)
  Clustered rows: 955,355
```

3. Monthly Aggregation

Parse `RECVDATE` into datetime, restrict records to the study window (2020–2025), and aggregate to monthly totals.

- **`MONTHLY_REPORTS`** — total number of adverse event reports received that month
- **`SEVERE_PROPORTION`** — proportion of those reports classified as serious (`SERIOUS = 1`)

In [ ]:
```
# TIME SERIES ANALYSIS: MONTHLY REPORTING TRENDS (2020-2025)

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm


# 1. Ensure RECVDATE is datetime
df_ts = df_clean.copy()

df_ts["RECVDATE"] = pd.to_datetime(df_ts["RECVDATE"], errors="coerce")

# Keep only records with valid RECVDATE
df_ts = df_ts[df_ts["RECVDATE"].notna()].copy()
```

```python
# Restrict to study period 2020-2025
df_ts = df_ts[
    (df_ts["RECVDATE"] >= "2020-01-01") &
    (df_ts["RECVDATE"] < "2026-01-01")
].copy()

print("Time series dataset shape:", df_ts.shape)


# 2. Create year-month variable
df_ts["YEAR_MONTH"] = df_ts["RECVDATE"].dt.to_period("M").astype(str)

# Also keep a sortable datetime version for plotting/regression
df_ts["YEAR_MONTH_DATE"] = df_ts["RECVDATE"].dt.to_period("M").dt.to_timesta


# 3. Monthly reporting volume + severe proportion
monthly_summary = (
    df_ts.groupby("YEAR_MONTH_DATE")
    .agg(
        MONTHLY_REPORTS=("VAERS_ID", "count"),
        SEVERE_CASES=("SERIOUS", "sum")
    )
    .reset_index()
)

monthly_summary["SEVERE_PROPORTION"] = (
    monthly_summary["SEVERE_CASES"] / monthly_summary["MONTHLY_REPORTS"]
)

# Create numeric time index for regression
monthly_summary = monthly_summary.sort_values("YEAR_MONTH_DATE").reset_index
monthly_summary["MONTH_INDEX"] = np.arange(len(monthly_summary))

print(monthly_summary.head())

# Save summary
monthly_summary.to_csv(f"{OUTDIR}/monthly_reporting_summary_2020_2025.csv",
```

```
Time series dataset shape: (986096, 106)
  YEAR_MONTH_DATE  MONTHLY_REPORTS  SEVERE_CASES  SEVERE_PROPORTION  \
0      2020-12-01            10025          1800           0.179551
1      2021-01-01            70093          8250           0.117701
2      2021-02-01            60075          8025           0.133583
3      2021-03-01            73980         12153           0.164274
4      2021-04-01           100372         18034           0.179672

   MONTH_INDEX
0            0
1            1
2            2
3            3
4            4
```

4. Reporting Volume Trend

Fit a linear regression of monthly report count by month to assess whether overall VAERS reporting volume changed significantly across the study period.

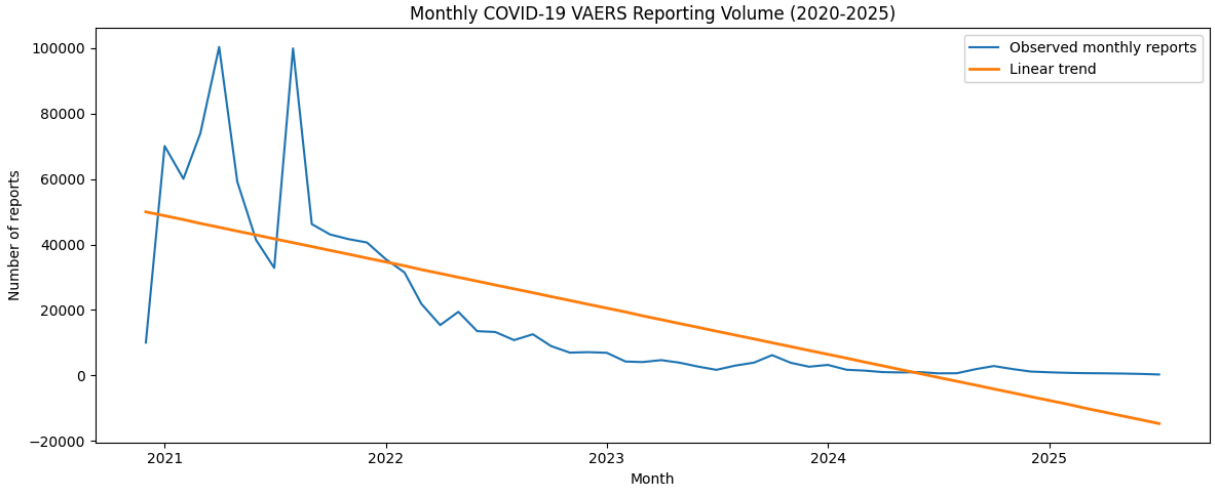
Model: `MONTHLY_REPORTS ~ MONTH_INDEX`

Coefficients, p-values, and 95% confidence intervals are saved to `Outputs/monthly_volume_regression_results.csv`. The observed series and fitted linear trend are plotted together.

In [ ]:
```
# ==============================================================
# LINEAR REGRESSION: MONTHLY REPORTING VOLUME OVER TIME
# ==============================================================

# Model 1: Monthly reporting volume ~ time
X_volume = sm.add_constant(monthly_summary["MONTH_INDEX"])
y_volume = monthly_summary["MONTHLY_REPORTS"]

volume_model = sm.OLS(y_volume, X_volume).fit(cov_type="HC3")

print("MONTHLY REPORTING VOLUME MODEL")
print(volume_model.summary())

# Save model coefficients
volume_results = pd.DataFrame({
    "term": volume_model.params.index,
    "coefficient": volume_model.params.values,
    "p_value": volume_model.pvalues.values,
    "conf_low": volume_model.conf_int()[0].values,
    "conf_high": volume_model.conf_int()[1].values
})
volume_results.to_csv(f"{OUTDIR}/monthly_volume_regression_results.csv", inc

# Plot monthly volume with fitted regression line
monthly_summary["VOLUME_PRED"] = volume_model.predict(X_volume)

plt.figure(figsize=(12, 5))
plt.plot(monthly_summary["YEAR_MONTH_DATE"], monthly_summary["MONTHLY_REPORT
plt.plot(monthly_summary["YEAR_MONTH_DATE"], monthly_summary["VOLUME_PRED"],
plt.title("Monthly COVID-19 VAERS Reporting Volume (2020-2025)")
plt.xlabel("Month")
plt.ylabel("Number of reports")
plt.legend()
plt.tight_layout()
plt.show()
```

MONTHLY REPORTING VOLUME MODEL

```
                            OLS Regression Results
==============================================================================
==
Dep. Variable:        MONTHLY_REPORTS   R-squared:                       0.5
81
Model:                            OLS   Adj. R-squared:                  0.5
73
Method:                 Least Squares   F-statistic:                     42.
89
Date:                Sat, 14 Mar 2026   Prob (F-statistic):           2.22e-
08
Time:                        13:35:00   Log-Likelihood:                -622.
08
No. Observations:                  56   AIC:                             124
8.
Df Residuals:                      54   BIC:                             125
2.
Df Model:                           1
Covariance Type:                  HC3
==============================================================================
===
                  coef    std err          z      P>|z|      [0.025      0.9
75]
------------------------------------------------------------------------------
---
const        4.996e+04   6737.736      7.415      0.000    3.68e+04    6.32e
+04
MONTH_INDEX -1176.5016    179.652     -6.549      0.000   -1528.613    -824.
390
==============================================================================
==
Omnibus:                       24.791   Durbin-Watson:                   0.9
87
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               50.3
97
Skew:                           1.340   Prob(JB):                     1.14e-
11
Kurtosis:                       6.797   Cond. No.                         6
3.0
==============================================================================
==

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```



5. Severity Proportion Trend

Fit a second linear regression to evaluate whether the monthly proportion of serious reports changed over time, independent of overall volume.

Model: `SEVERE_PROPORTION ~ MONTH_INDEX`

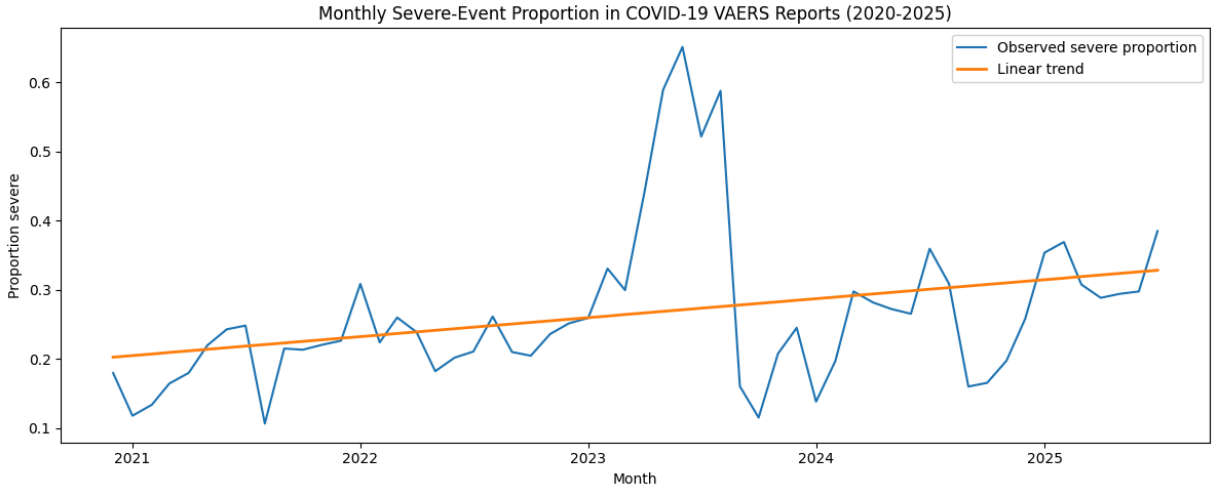
Coefficients, p-values, and confidence intervals are saved to `Outputs/monthly_severe_proportion_regression_results.csv`. The observed series and fitted trend are plotted together.

In [ ]:

```
# ============================================================
# LINEAR REGRESSION: MONTHLY SEVERE-EVENT PROPORTION OVER TIME
# ============================================================

# Model 2: Monthly severe proportion ~ time
X_severe = sm.add_constant(monthly_summary["MONTH_INDEX"])
y_severe = monthly_summary["SEVERE_PROPORTION"]

severe_model = sm.OLS(y_severe, X_severe).fit(cov_type="HC3")

print("MONTHLY SEVERE PROPORTION MODEL")
print(severe_model.summary())

# Save model coefficients
severe_results = pd.DataFrame({
    "term": severe_model.params.index,
    "coefficient": severe_model.params.values,
    "p_value": severe_model.pvalues.values,
    "conf_low": severe_model.conf_int()[0].values,
    "conf_high": severe_model.conf_int()[1].values
})
severe_results.to_csv(f"{OUTDIR}/monthly_severe_proportion_regression_result

# Plot monthly severe-event proportion with fitted regression line
monthly_summary["SEVERE_PROP_PRED"] = severe_model.predict(X_severe)
```

```
plt.figure(figsize=(12, 5))
plt.plot(monthly_summary["YEAR_MONTH_DATE"], monthly_summary["SEVERE_PROPORT
plt.plot(monthly_summary["YEAR_MONTH_DATE"], monthly_summary["SEVERE_PROP_PR
plt.title("Monthly Severe-Event Proportion in COVID-19 VAERS Reports (2020-2
plt.xlabel("Month")
plt.ylabel("Proportion severe")
plt.legend()
plt.tight_layout()
plt.show()
```

```
MONTHLY SEVERE PROPORTION MODEL
                            OLS Regression Results
==============================================================================
==
Dep. Variable:      SEVERE_PROPORTION   R-squared:                       0.1
06
Model:                            OLS   Adj. R-squared:                  0.0
90
Method:                 Least Squares   F-statistic:                     18.
65
Date:                Sat, 14 Mar 2026   Prob (F-statistic):           6.77e-
05
Time:                        13:36:21   Log-Likelihood:                 45.5
87
No. Observations:                  56   AIC:                            -87.
17
Df Residuals:                      54   BIC:                            -83.
12
Df Model:                           1
Covariance Type:                  HC3
==============================================================================
===
                 coef    std err          z      P>|z|      [0.025      0.9
75]
------------------------------------------------------------------------------
---
const          0.2024      0.016     12.371      0.000       0.170       0.
234
MONTH_INDEX    0.0023      0.001      4.318      0.000       0.001       0.
003
==============================================================================
==
Omnibus:                       29.853   Durbin-Watson:                   0.6
49
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               57.5
22
Skew:                           1.722   Prob(JB):                     3.23e-
13
Kurtosis:                       6.576   Cond. No.                         6
3.0
==============================================================================
==

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```



6. Cluster Prevalence Over Time

For each calendar month, compute what share of that month's clustered reports belonged to each narrative cluster. This shows whether the mix of adverse-event phenotypes shifted across phases of the COVID-19 vaccine rollout.

Steps:

1. Filter to rows that received a cluster label (i.e. had ≥ 3 tokens after cleaning)
2. Count reports per `(YEAR_MONTH_DATE, CLUSTER)` combination
3. Divide by the total clustered reports in that month to obtain the monthly proportion for each cluster
4. Save `Outputs/monthly_cluster_proportions.csv`

In [ ]:
```
# ============================================================
# MONTHLY CLUSTER PREVALENCE OVER TIME
# ============================================================

# Make sure df_clean has CLUSTER column
if "CLUSTER" not in df_clean.columns:
    print("CLUSTER column not found in df_clean")
else:
    # Start fresh from df_clean
    df_ts = df_clean.copy()

    # Ensure RECVDATE is datetime
    df_ts["RECVDATE"] = pd.to_datetime(df_ts["RECVDATE"], errors="coerce")

    # Create YEAR_MONTH_DATE BEFORE filtering
    df_ts["YEAR_MONTH_DATE"] = df_ts["RECVDATE"].dt.to_period("M").dt.to_tim

    # Now filter to rows with cluster labels
    df_cluster_ts = df_ts[df_ts["CLUSTER"].notna()].copy()
    df_cluster_ts["CLUSTER"] = df_cluster_ts["CLUSTER"].astype(int)

    print("Rows with cluster labels:", df_cluster_ts.shape[0])
```

```
    print("Unique clusters:", sorted(df_cluster_ts["CLUSTER"].unique()))
    print("Date range:", df_cluster_ts["YEAR_MONTH_DATE"].min(), "to", df_cl

    # Monthly counts by cluster
    cluster_month_counts = (
        df_cluster_ts.groupby(["YEAR_MONTH_DATE", "CLUSTER"])
        .size()
        .reset_index(name="N_CLUSTER_MONTH")
    )

    # Total clustered reports per month
    cluster_month_totals = (
        df_cluster_ts.groupby("YEAR_MONTH_DATE")
        .size()
        .reset_index(name="N_TOTAL_MONTH")
    )

    # Merge totals and compute proportions
    cluster_month_props = cluster_month_counts.merge(
        cluster_month_totals,
        on="YEAR_MONTH_DATE",
        how="left"
    )

    cluster_month_props["CLUSTER_PROPORTION"] = (
        cluster_month_props["N_CLUSTER_MONTH"] / cluster_month_props["N_TOTA
    )

    # Save
    cluster_month_props.to_csv(f"{OUTDIR}/monthly_cluster_proportions.csv",

    print("\nCluster-month proportions:")
    print(cluster_month_props.head(10))
```

```
Rows with cluster labels: 955355
Unique clusters: [np.int64(0), np.int64(1), np.int64(2), np.int64(3), np.int
64(4), np.int64(5), np.int64(6), np.int64(7)]
Date range: 2020-12-01 00:00:00 to 2025-07-01 00:00:00

Cluster-month proportions:
  YEAR_MONTH_DATE  CLUSTER  N_CLUSTER_MONTH  N_TOTAL_MONTH  CLUSTER_PROPORTI
ON
0      2020-12-01        0               27           9731            0.0027
75
1      2020-12-01        1             2591           9731            0.2662
62
2      2020-12-01        2              313           9731            0.0321
65
3      2020-12-01        3             2212           9731            0.2273
15
4      2020-12-01        4              576           9731            0.0591
92
5      2020-12-01        5               52           9731            0.0053
44
6      2020-12-01        6             3767           9731            0.3871
13
7      2020-12-01        7              193           9731            0.0198
34
8      2021-01-01        0              497          68152            0.0072
93
9      2021-01-01        1             9441          68152            0.1385
29
```

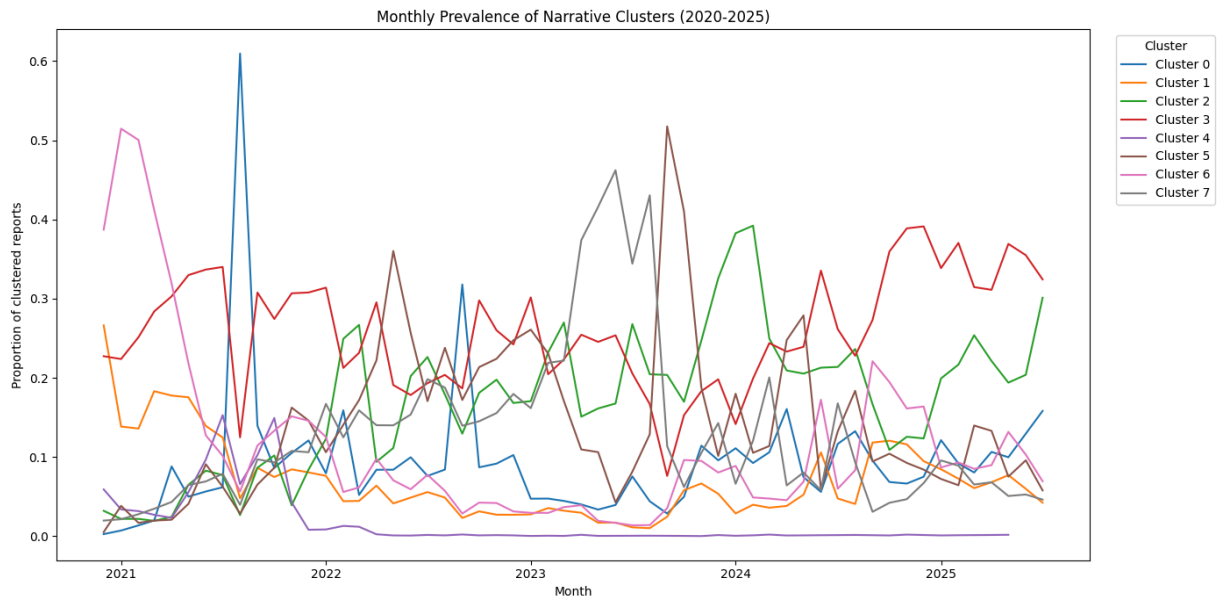
7. Plot Cluster Prevalence

Line plot showing each cluster's monthly share of reports from January 2020 through March 2025. Each of the 8 clusters is rendered as a separate line, with the legend placed outside the axes to prevent overlap.

In [ ]:

```
# ============================================================
# PLOT CLUSTER PREVALENCE OVER TIME
# ============================================================

plt.figure(figsize=(14, 7))

for cluster_id in sorted(cluster_month_props["CLUSTER"].unique()):
    cluster_subset = cluster_month_props[cluster_month_props["CLUSTER"] == c
    plt.plot(
        cluster_subset["YEAR_MONTH_DATE"],
        cluster_subset["CLUSTER_PROPORTION"],
        label=f"Cluster {cluster_id}"
    )

plt.title("Monthly Prevalence of Narrative Clusters (2020-2025)")
plt.xlabel("Month")
plt.ylabel("Proportion of clustered reports")
plt.legend(title="Cluster", bbox_to_anchor=(1.02, 1), loc="upper left")
```

```
plt.tight_layout()
plt.show()
```



8. Cluster Trend Regressions

For each of the 8 narrative clusters, fit an individual linear regression.

Results saved to `Outputs/cluster_trend_regression_results.csv`.

In [ ]:
```
# ============================================================
# LINEAR REGRESSION FOR EACH CLUSTER'S MONTHLY PREVALENCE
# ============================================================

cluster_trend_results = []

for cluster_id in sorted(cluster_month_props["CLUSTER"].unique()):
    cluster_subset = (
        cluster_month_props[cluster_month_props["CLUSTER"] == cluster_id]
        .sort_values("YEAR_MONTH_DATE")
        .reset_index(drop=True)
    )

    cluster_subset["MONTH_INDEX"] = np.arange(len(cluster_subset))

    X_cluster = sm.add_constant(cluster_subset["MONTH_INDEX"])
    y_cluster = cluster_subset["CLUSTER_PROPORTION"]

    model = sm.OLS(y_cluster, X_cluster).fit(cov_type="HC3")

    cluster_trend_results.append({
        "CLUSTER": cluster_id,
        "INTERCEPT": model.params["const"],
        "SLOPE": model.params["MONTH_INDEX"],
        "P_VALUE": model.pvalues["MONTH_INDEX"],
        "R_SQUARED": model.rsquared
    })
```

```
cluster_trend_results_df = pd.DataFrame(cluster_trend_results).sort_values("
print(cluster_trend_results_df)

cluster_trend_results_df.to_csv(f"{OUTDIR}/cluster_trend_regression_results.
```

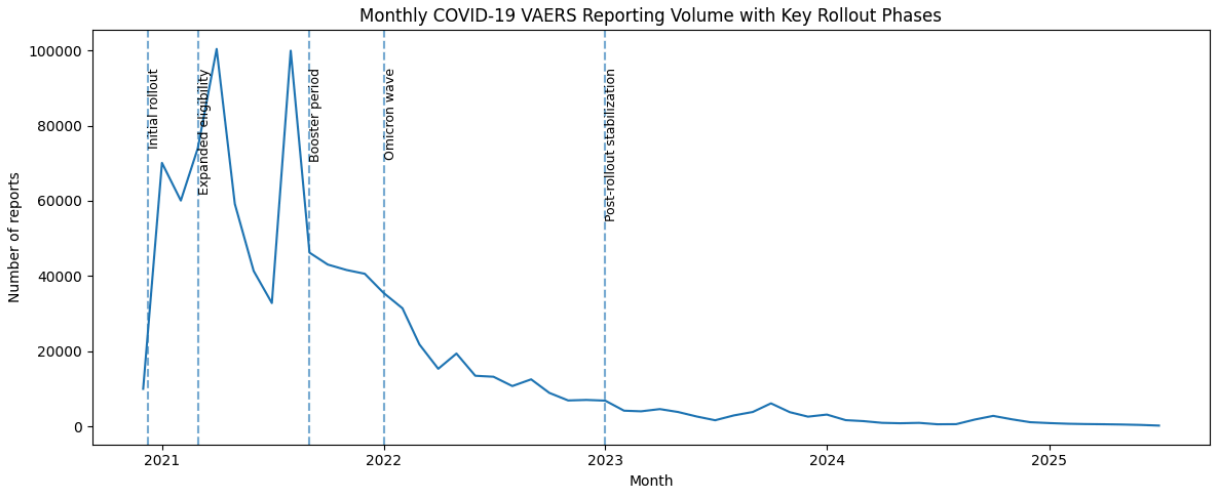
```
   CLUSTER  INTERCEPT     SLOPE       P_VALUE  R_SQUARED
2        2   0.075727  0.003481  1.586777e-10   0.410408
4        4   0.057977 -0.001705  6.237782e-06   0.347469
1        1   0.103921 -0.001221  1.323087e-02   0.155424
6        6   0.193502 -0.002845  1.540750e-02   0.170204
3        3   0.234100  0.000946  9.672188e-02   0.048129
5        5   0.122053  0.000739  3.098287e-01   0.014494
0        0   0.092119  0.000083  9.201333e-01   0.000251
7        7   0.128494  0.000039  9.513804e-01   0.000039
```

9. Annotated Timeline Plots

Re-plots monthly report volume with vertical dashed lines marking the five key phases of the U.S. COVID-19 vaccine programme:

Phase marker	Date
Initial rollout begins	December 2020
Expanded eligibility (all adults)	March 2021
Booster authorisation period	September 2021
Omicron wave peak	January 2022
Post-rollout stabilisation	January 2023

In [ ]:

```
# ============================================================
# ANNOTATED PLOTS FOR KEY ROLLOUT PHASES
# ============================================================

# You can customize these dates depending on what phases
# you want to highlight in your report.
rollout_dates = [
    ("2020-12-08", "Initial rollout"),
    ("2021-03-01", "Expanded eligibility"),
    ("2021-09-01", "Booster period"),
    ("2022-01-01", "Omicron wave"),
    ("2023-01-01", "Post-rollout stabilization")
]

# Monthly volume plot with annotations
plt.figure(figsize=(12, 5))
plt.plot(monthly_summary["YEAR_MONTH_DATE"], monthly_summary["MONTHLY_REPORT

for date_str, label in rollout_dates:
    date_val = pd.to_datetime(date_str)
    plt.axvline(date_val, linestyle="--", alpha=0.6)
    plt.text(date_val, monthly_summary["MONTHLY_REPORTS"].max()*0.95, label,
```

```
                rotation=90, verticalalignment="top", fontsize=9)

plt.title("Monthly COVID-19 VAERS Reporting Volume with Key Rollout Phases")
plt.xlabel("Month")
plt.ylabel("Number of reports")
plt.tight_layout()
plt.show()
```